



CHAPTER

Installation

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4.1 Delivery and unloading

TODO: no document describing crane/forklift unloading, container weight, or lifting-point locations was found. The container general-arrangement drawings (2308CA1/2308CA2, appendix B) likely show lifting points but have not yet been reviewed for this purpose.



Figure 4.1. Suggested figure: photograph or diagram of the container being lifted/unloaded on site, with lifting points marked, once the procedure is confirmed.

4.2 Site requirements

The system is built around one or two ISO shipping containers (pretreatment + treatment, chapter 3), 20 ft/40 ft, High-Cube or Standard. In HydroVolta's documented BASF pilot, the pretreatment unit used a 10 ft container. The site must provide a water source (the pilot drew from a cooling-tower basin via a submersible pump) and space alongside the containers for a drain tank. TODO: ambient temperature range, ground/ foundation/leveling requirements and clearances beyond what the pilot layout implies were not found in a dedicated site-requirements document.

4.3 Positioning the container

TODO: no explicit positioning/orientation document was found. From the piping documented for commissioning (section 4.4), the pretreatment container's filtered-water outlet (2") must be within pipe-run distance of the treatment container's inlet (2"), and both containers' 3" drain outlets should be positioned so they can be tee'd together to a shared drain tank.

4.4 Water connections

- **Pretreatment container:** feed inlet 2½" (M) from the raw water source (in the pilot, a submersible pump rated 4.5 m³/h at 6 bar); filtered-water outlet 2" (M); drain outlet 3" (M).
- **Treatment container:** inlet 2" (M), fed from the pretreatment filtered-water outlet; product-water outlet 2" (M); drain outlet 3" (M).
- Pump-to-pretreatment feed line: PVCu, 60 mm, terminated 2½" (F).
- Pretreatment-to-treatment line: PVCu, 40 mm.
- Both containers' 3" (M) drain outlets combine via a tee into one line to the shared drain tank.
- Fit O-rings and PTFE tape at every threaded joint.

Key in-line pretreatment components: an ultrafiltration unit (AKKIM APM 80) via 2" Victaulic couplings, and disc filters (AZUD DF100/DF50/DF20) via 2" (F) 63 mm transition pieces.

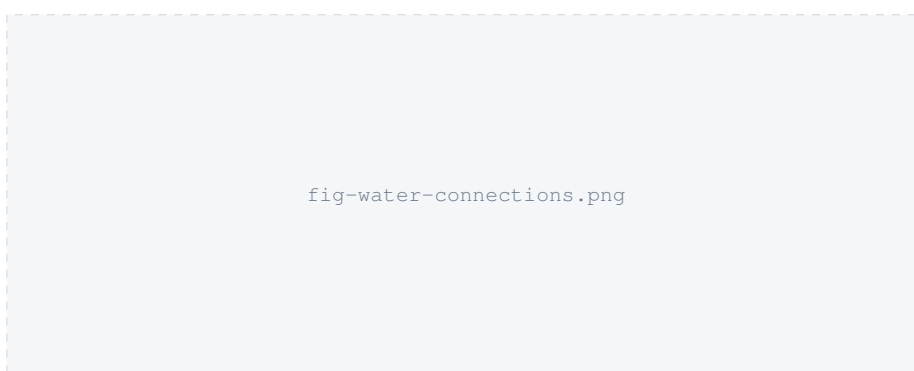


Figure 4.2. Suggested figure: a labeled diagram of the pretreatment/treatment container pair showing every hydraulic connection point (feed, filtered-water, product, drain) with its size and thread type.

4.5 Electrical connection

The control cabinet is IP55-rated. Incoming supply on the documented pilot was 3×400 V AC (3P+N+PE, per IEC 60445), max. 25 A, with a 24 V DC control bus for valves and I/O (from the pretreatment electrical schematic, sheet E1/document I2). A separate electrical schematic series exists specifically for a **treatment container** (as opposed to pretreatment) under the BASF pilot project (folder `2_Electrical/ 0_Treatment container`, multiple dated AutoCAD Electrical `.dsn` revisions) — TODO: this series was located but not opened (CAD format); a human with AutoCAD Electrical should open the latest revision for the treatment-container-specific breaker/RCD sizing, since the figures above are from the pretreatment side only. TODO: confirm whether either value set applies to the standard SalinBloc™ product or is pilot-specific.

4.6 Network and telemetry connection

See chapter 16 for the router, remote-access gateway and cloud dashboard used for network/telemetry connectivity.

4.7 Installation checklist

1. Confirm all hydraulic connections listed in section 4.4 are complete: pretreatment feed/outlet/drain, treatment inlet/outlet/drain, combined drain-to-tank tee.
2. Confirm electrical supply matches the cabinet's rated voltage/current (chapter 14) before energizing.
3. Confirm network/telemetry connection is established (chapter 16).

TODO: no standalone installation checklist document was found; the list above is assembled from the commissioning data sheet and should be reviewed against a real site installation before publishing.